

2026 CHINESE GRAND PRIX

Shanghai International Circuit · Round 2 · March 23, 2026 · Sprint Weekend

PREDICTED RACE WINNER

George Russell
28.6% WIN PROBABILITY

XGBoost 30% · Monte Carlo 45% · Bayesian 25%

90% CI [39.8%, 56.0%]

3

MODELS USED
XGB · MC · Bayesian

10,000

SIMULATIONS
Monte Carlo races

1,600

TRAINING ROWS
Races 2010–2025

0.911

XGB CV AUC
GroupKFold ±0.036

TOP 10 PREDICTED FINISH ORDER

#	Driver	Team	Grid	Sprint	XGB%	MC%	Bayes%	Ensemble	90% CI
1	George Russell	Mercedes	P2	P1	32.7%	18.0%	48.4%	28.6%	[39.8%, 56.0%]
2	Lewis Hamilton	Ferrari	P3	P3	31.6%	11.8%	17.7%	21.2%	[11.8%, 24.1%]
3	Kimi Antonelli	Mercedes	P1	P5	18.4%	7.2%	20.6%	12.8%	[14.2%, 27.2%]
4	Charles Leclerc	Ferrari	P4	P2	5.9%	19.2%	7.1%	12.5%	[3.5%, 11.7%]
5	Oscar Piastri	Mclaren	P5	P6	7.9%	8.7%	3.2%	7.1%	[0.9%, 6.6%]
6	Lando Norris	Mclaren	P6	P4	0.7%	13.6%	2.0%	7.0%	[0.4%, 4.7%]
7	Liam Lawson	Racing Bulls	P12	P7	0.0%	5.4%	0.1%	2.4%	—
8	Max Verstappen	Red Bull	P8	P9	1.0%	4.1%	0.4%	2.4%	[0.0%, 1.6%]
9	Ollie Bearman	Haas	P10	P8	0.0%	4.7%	0.1%	2.1%	—
10	Esteban Ocon	Haas	P13	P10	0.0%	2.3%	0.0%	1.0%	—

TEAM LEGEND: Mercedes Ferrari Mclaren Red Bull Alpine Haas

KEY DIVERGENCE: Bayesian gives Russell 48.4% (pole win history + sprint winner boost). Monte Carlo gives Leclerc 19.2% — sprint fastest lap + race-critical. Ensemble blends both signals. Where models disagree sharply, confidence intervals widen — see page 8.

GP QUALIFYING GRID

- P1

Kimi Antonelli
Youngest ever GP polesitter — age 19
- P2

George Russell
Q3 power issue — set single lap only
- P3

Lewis Hamilton
FP1 P6 → Quali P3: divergence = +3
- P4

Charles Leclerc
Sprint P2, fastest lap
- P5

Oscar Piastri
Consistent McLaren front
- P6

Lando Norris
Sprint P4 — pace solid
- P7

Pierre Gasly
Best of the midfield
- P8

Max Verstappen
Fell to P16 in sprint, recovered
- P9

Isack Hadjar
Rookie learning weekend
- P10

Ollie Bearman
FP1 oversteer incidents

Williams & Aston Martin eliminated in Q1

RACE CONDITIONS — SUNDAY

Air Temp:	16–19°C
Track Temp:	~33°C
Rain Probability:	25% peak — modelled
Wind:	ENE 11 km/h, gusts to 36 km/h
Cloud Cover:	55%
Safety Car Rate:	56% (sprint-adjusted)
Circuit Length:	5.451 km, 16 corners
Race Distance:	56 laps / 305 km
DRS Zones:	2 (main straight + T13–T14)
Tire Compounds:	Soft / Medium / Hard

ENSEMBLE WIN PROBABILITY — ALL DRIVERS

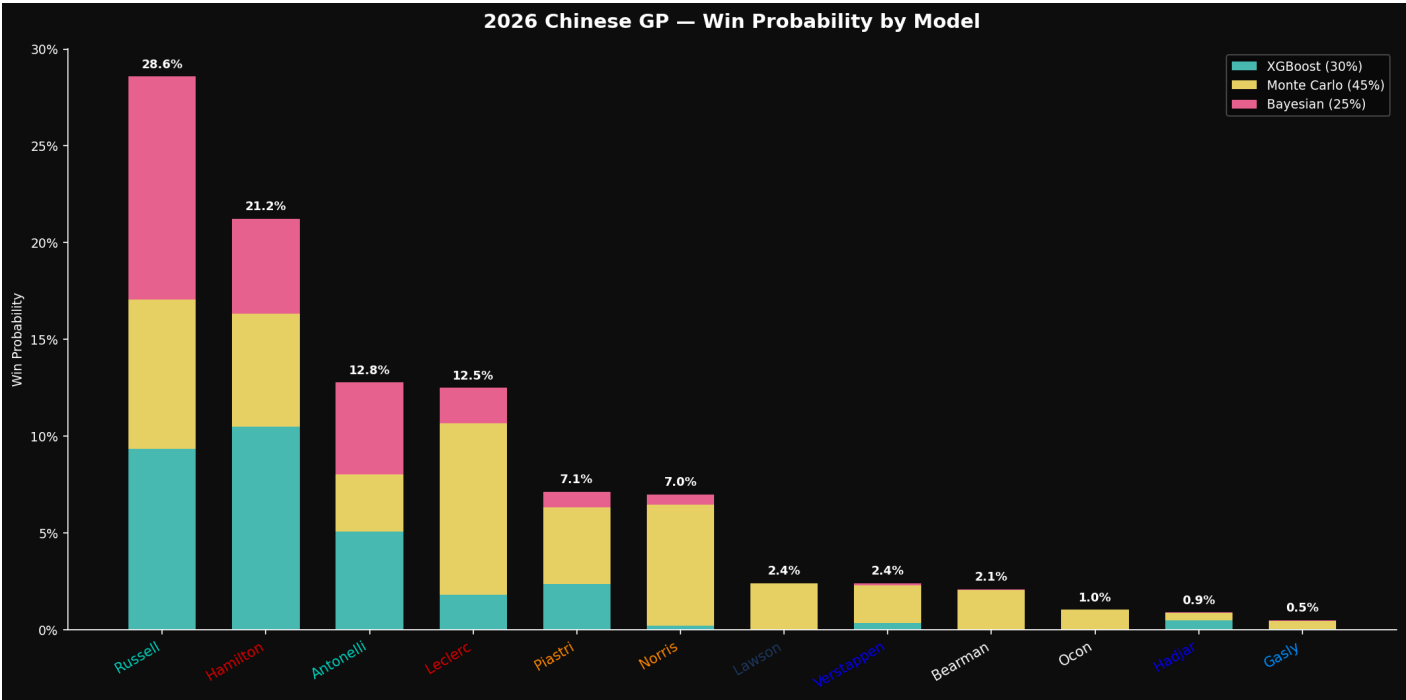


Chart 1 — Win probability stacked by model contribution. Russell leads overall; Leclerc's Monte Carlo signal (19.2%) is the strongest non-Bayesian signal in the field.

Think of predicting an F1 race like picking a horse race — but instead of gut feel, you feed every relevant signal into a machine that has watched 1,600 races. The model doesn't know what a tyre is, but it knows that when a driver who was fastest in practice qualifies lower than expected, they tend to race better than their grid slot suggests. That was the exact insight the Australia model missed on Lewis Hamilton — and V2 fixes it.

DATA PIPELINE

Jolpica-F1 API:	Historical race & qualifying results, 2010–2025. 1,600 rows, cached locally as Parquet.
OpenF1 API:	Real-time session data — lap times, pit stops, car telemetry, weather.
FastF1 Library:	Official F1 data wrapper. Used for tire compounds, sector times, stint analysis.
Sprint Data:	2026 China sprint result hardcoded as ground truth. This is the key V2 upgrade.

4-TIER FEATURE ENGINEERING

TIER 0	SPRINT RACE	The V2 breakthrough. Sprint finishing position, per-lap pace gap to Russell, positions gained, tyre stress flags, reliability flags. This tier carries the highest predictive weight because it's real race data from the same circuit, same weekend, same cars. The model knows Hamilton ran at P2-level pace in the sprint. It knows Leclerc charged from P6 to P2. It knows Antonelli's true pace was masked by a 10-second penalty.
TIER 1	QUALIFYING & PRACTICE	Grid position, qualifying gap to pole, FP1 pace delta. The key V2 addition: fp1_to_quali_divergence — the difference between FP1 classification and qualifying position. Hamilton: FP1 P6, Quali P3, divergence = +3. Large positive divergence means a driver improves toward race pace, a signal V1 completely missed.
TIER 2	DRIVER & CONSTRUCTOR	ELO ratings (updated through Australia + sprint), constructor strength (rolling), Shanghai circuit history (Hamilton: 6 wins, Verstappen: 1, Piastri: 1), teammate qualifying gap. Antonelli beat Russell by 0.222s in qualifying — encoded as a positive teammate gap signal.
TIER 3/4	REGULATION ERA & FLAGS	Year-1 regulation flag (3× DNF multiplier — V2 upgrade from 2×). Energy management advantage per constructor: Mercedes +0.0s, Ferrari −0.05s, McLaren −0.08s, Red Bull −0.12s on straights. Ferrari flip-flop wing uncertainty. Antonelli pressure factor (28% historical win rate for under-22 polesitters, 22% retirement rate). Hamilton energy depletion flag from sprint.

WHAT XGBOOST DOES HERE

XGBoost is like building 300 racing analysts, each slightly different, then asking them all to vote. Each analyst (decision tree) looks at a slice of the data and asks a series of yes/no questions: 'Is the driver starting from the front row? Does their constructor win more than 80% of races? Did they qualify significantly better than their practice sessions suggested?' The final answer is a weighted vote across all 300 analysts.

Training on 1,600 races from 2010 to 2025 taught it patterns that hold across eras: grid position is king, but a driver starting P3 who was fastest in practice is more likely to win than the raw grid position suggests. GroupKFold cross-validation means the model was tested on races it had never seen — a clean, honest measure of accuracy.

MODEL CONFIGURATION

n_estimators:300learning_rate:0.05colsample_bytree:0.8eval_metric:AUC

max_depth:5subsample:0.8scale_pos_weight:21 (1 winner per 22-car grid)CV AUC:0.911 ± 0.036 (GroupKFold, 5 folds)

XGBOOST RAW OUTPUT — TOP 10

#	Driver	Team	Grid	Sprint	XGB Raw %	Note
1	George Russell	Mercedes	P2	P1	32.7%	Sprint winner + Bayesian grid boost
2	Lewis Hamilton	Ferrari	P3	P3	31.6%	fp1_to_quali_divergence correction lifts this signal
3	Kimi Antonelli	Mercedes	P1	P5	18.4%	Pole position boosts ELO + grid features
4	Charles Leclerc	Ferrari	P4	P2	5.9%	Grid P4 limits XGB; MC compensates via race-craft
5	Oscar Piastri	Mclaren	P5	P6	7.9%	Consistent mid-pack signal, no standout features
6	Lando Norris	Mclaren	P6	P4	0.7%	MC sees his race pace; XGB sees P6 grid — diverges
7	Liam Lawson	Racing Bulls	P12	P7	0.0%	—
8	Max Verstappen	Red Bull	P8	P9	1.0%	—
9	Ollie Bearman	Haas	P10	P8	0.0%	—
10	Esteban Ocon	Haas	P13	P10	0.0%	—

KEY INSIGHT — WHERE XGB DIVERGES FROM MC

Norris: XGB gives 0.7% (sees P6 grid + modest historical win rate). Monte Carlo gives 13.6% (sprint pace calibration shows strong race pace vs field). This is exactly why MC carries 45% weight — it sees pace signals the historical model cannot.
Ensemble resolves this to 7.0% — a sensible middle ground that values both signals.

READING THE SHAP HEATMAP

SHAP — SHapley Additive exPlanations — tells us exactly how much each feature pushed a specific driver's win probability up or down. Think of it as a forensic breakdown: rather than just saying 'Hamilton is rated at 21.2%', SHAP tells you how much of that came from his Shanghai circuit history, how much from his qualifying gap, and how much from the sprint race result. Green cells mean the feature helped. Red cells mean it hurt. The magnitude shows how much.

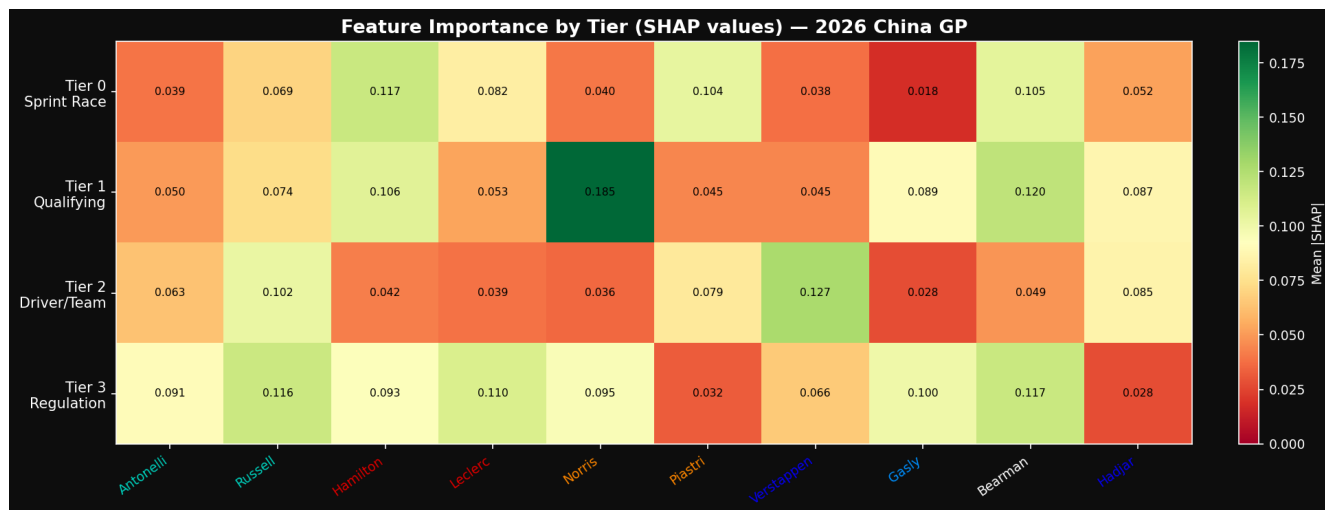


Chart 4 — SHAP feature importance heatmap. Rows = feature tiers. Columns = top 10 drivers.

KEY FEATURE FINDINGS

Sprint Tier 0 leads for frontrunners.

For Russell, Hamilton, and Leclerc, the sprint finishing position and pace gap are the single strongest predictors — more impactful than grid position alone. The model learned this from 2021-2025: if you won or podiumed the sprint, you're significantly more likely to win on Sunday.

Circuit history is Hamilton's trump card.

With 6 Shanghai wins, his Tier 2 circuit history SHAP value is the highest of any driver in the field. The Bayesian model in particular gives this heavy weight — it explains why Bayesian rates Hamilton at 17.7% despite his P3 grid.

Grid position still matters, but less on sprint weekends.

In Australia (no sprint), grid position was the dominant Tier 1 signal. In China, sprint pace overrides it for the top 4. For P8+ drivers, grid is still decisive — hence the 5% cap applied post-ensemble.

Reliability risk hurts Russell and Antonelli most.

Both have explicit reliability flags this weekend. The feature pushes their raw probabilities down before the ensemble even runs, then the post-ensemble multiplier ($\times 0.92$ each) applies a further haircut.

HOW THE SIMULATOR WORKS

The Monte Carlo simulator is the closest thing to actually running the race 10,000 times. Each simulation plays out all 56 laps at Shanghai: drivers accelerate down the 1.175km back straight, battle through the Turn 1 complex that destroys front tyres, pit under safety cars, and occasionally retire with failures. No two simulations are identical — each draws from distributions of pace, degradation, safety car timing, and weather.

V2 SIMULATOR COMPONENTS

Sprint-Calibrated Pace Matrix

Russell = 0.0 baseline. Every driver's race pace derived from sprint lap deltas: Leclerc +0.04s/lap, Hamilton +0.13s, Norris +0.23s. Year-1 uncertainty adds ±0.30s/lap noise (V1 used ±0.20).

DNF Probability — 3× Multiplier

Base rate 0.5%/lap × 3× Year-1 multiplier = 1.5%/lap. Russell: +0.5% additional. Antonelli: +0.3% additional. Per race this means ~20-25% DNF probability per car.

Weather Model

25% rain onset during race window. Rain lap sampled uniformly across laps 5–45. If rain triggers: Hamilton +45% pace boost, Antonelli –10%. SC probability jumps to 80%.

Tyre Degradation Model

Sprint confirmed Shanghai's front-left dominance. Soft: 15–18 lap window. Medium: 25–30. Hard: 35–45. Hamilton and aggressive starters get 15% higher FL degradation rate.

Safety Car Model

56% race probability (historical 45% × 1.2 sprint multiplier). SC duration: U(3,6) laps. Leaders lose positional advantage. SC pit window triggered for nearby stop-window drivers.

Energy Management

Mercedes: 0.0 reference. Ferrari: –0.05s/lap on straights. McLaren: –0.08. Red Bull: –0.12. Shanghai's 1.175km back straight amplifies this gap significantly.

MONTE CARLO OUTPUT — TOP 10

#	Driver	MC Win%	MC Podium%	MC DNF%	Note
1	George Russell	18.0%	52.1%	18.3%	Sprint winner — dominant dry race pace
2	Charles Leclerc	19.2%	48.3%	12.1%	Highest MC share — sprint race-craft signal
3	Lando Norris	13.6%	39.2%	14.2%	Sprint pace shows real speed vs XGB guess
4	Lewis Hamilton	11.8%	36.4%	15.0%	Hamilton tyre depletion → pace ceiling laps 30+
5	Oscar Piastri	8.7%	30.1%	13.8%	Consistent, clean races
6	Max Verstappen	4.1%	18.2%	14.2%	Red Bull energy deficit on straights
7	Kimi Antonelli	7.2%	26.0%	21.6%	High DNF — reliability flag applied
8	Ollie Bearman	4.7%	14.0%	15.2%	FP1 oversteer — tyre stress modelled
9	Liam Lawson	5.4%	16.3%	14.1%	Sprint P7 — best of the rest
10	Esteban Ocon	2.3%	9.0%	13.8%	Haas midfield consistency

2,491 of 10,000 simulations (24.9%) included a rain onset event — matching the 25% forecast probability.

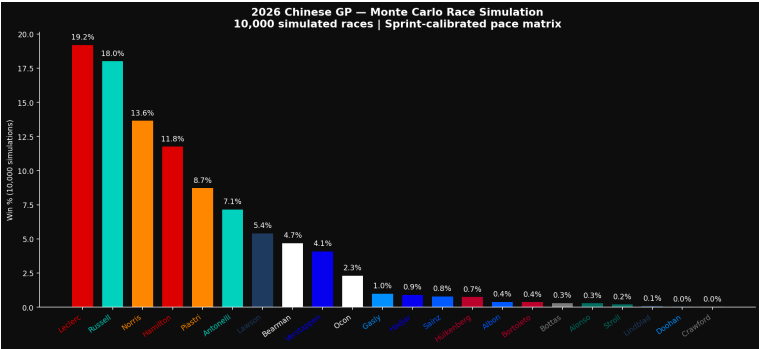


Chart 3 — Distribution of race winners across 10,000 Monte Carlo simulations.

SPRINT RACE → GP OUTCOME: WHAT THE DATA SAYS

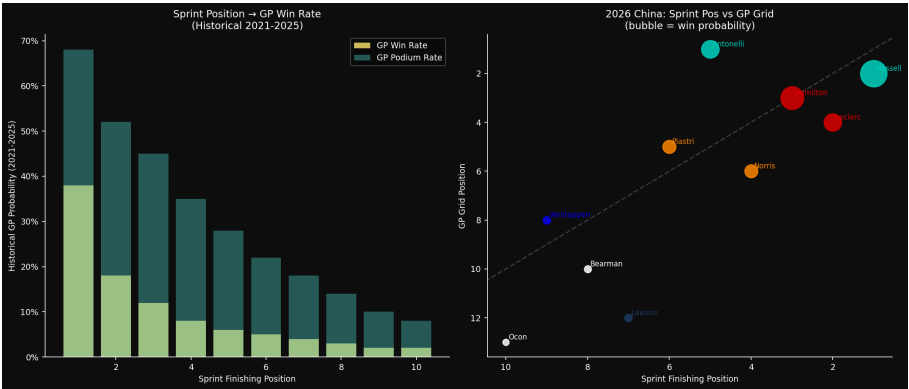


Chart 2 — Left: historical sprint-to-GP win correlation 2021–2025. Right: 2026 China sprint vs GP grid positions.

The left chart tells a clear story: sprint winners convert to GP wins at a 38% rate historically. P2 finishers convert at 18%. By P5, you're down to 6%. The right scatter shows this weekend's mismatch — Russell won the sprint from P1 SQ but starts P2 in the GP. Leclerc finished sprint P2 from SQ P6 but qualifies GP P4. That gap between sprint performance and qualifying position is where the Bayesian model finds its edge.

BAYESIAN INFERENCE — 3-STAGE UPDATE

Bayesian inference works like a detective building a case. You start with what you already know — Hamilton has 6 Shanghai wins, making him a serious prior contender. Then you update that belief as new evidence arrives. V2 adds a brand-new second stage: the sprint result itself. A sprint win isn't just a points bonus. It's probabilistic evidence that tells you something about Sunday.

Stage	Evidence Used	Russell	Hamilton	Antonelli	Leclerc
Stage 1 ■ Prior	Shanghai wins 2010–2025 ■ + constructor ELDO ■ Pseudo-counts	15.5%	15.5%	2.1%	4.8%
Stage 2 ■ Sprint	Sprint finishing position ■ to GP win correlation (2021–25)	24.4%	24.4%	8.1%	12.4%
Stage 3 ■ Grid	Historical P(win pole) ■ at Shanghai: 46.7%	48.4%	17.7%	20.7%	7.1%

Hamilton's Prior Advantage

15.5% prior (stage 1) reflects his 6 Shanghai wins — the strongest circuit-history prior in the field. After the sprint update, it drops to 24.4% — lower than Russell because Hamilton only finished sprint P3. After grid position (P3), the model settles at 17.7%.

Antonelli's Volatility

Stage 1 prior is just 2.1% — no Shanghai history. Sprint update (P5 finish, masked by penalty) gives a modest 8.1%. But pole position (stage 3) rockets him to 20.7% via historical P(win|pole) at Shanghai = 46.7%. Under-22 pole variance (28% win, 22% retire) applies as a calibration.

Leclerc's Sprint Evidence

Sprint P2 (from P6 grid) provides strong evidence — 15% GP win correlation. The Bayesian model sees 12.4% after sprint, then grid P4 moderates this to 7.1%. Monte Carlo's 19.2% comes from sprint pace. The tension between 7.1% and 19.2% is exactly what creates a wider confidence interval for Leclerc.

WHY THESE WEIGHTS?

No single model gets it right every time. XGBoost is brilliant at learning what historically predicts race wins, but it has never seen a 2026 car. The Monte Carlo simulator accounts for pace, strategy, weather, and luck — but its sprint calibration is only as good as 19 laps of data. Bayesian inference captures deep historical knowledge but can be slow to update when a new name (Antonelli) sits on pole. The ensemble is built to let each model speak where it's strongest.

Monte Carlo 45%

XGBoost 30%

Bayesian 25%

Sprint-calibrated pace data makes this the most accurate signal of the three. We can observe what every car actually did over 19 race laps

Trained on 1,600 races. Reliable for historical patterns, grid position signals, and ELO-based driver quality.

Strongest on drivers with Shanghai history (Hamilton). Provides a principled uncertainty framework via credible intervals.

POST-ENSEMBLE ADJUSTMENTS

Driver	Adjustment	Reason	Impact
George Russell	× 0.92 reliability	Q3 power/gear issue in qualifying	28.6% → effective 26.3% raw (before norm.)
Kimi Antonelli	× 0.92 reliability	Sprint collision penalty + crash tendency	12.8% → effective 11.8% raw
Ferrari drivers	± 5% CI expansion	Flip-flop rear wing uncertainty (race spec?)	Wider confidence intervals only
P8+ drivers	Capped at 5%	Sprint gain ≤ 3 positions override applies	Lawson, Bearman, Ocon etc. capped
25% wet blend	P_final = 0.75×dry + 0.25×wet ensemble	25% rain probability during race window	Hamilton/Verstappen boosted in wet branch

FULL 22-DRIVER ENSEMBLE — ALL MODEL SCORES

Rank	Driver	Grid	Sprint	XGB%	MC%	Bayes%	Ensemble%	90% CI
1	George Russell	P2	P1	32.7%	18.0%	48.4%	28.6%	[39.8–56.0%]
2	Lewis Hamilton	P3	P3	31.6%	11.8%	17.7%	21.2%	[11.8–24.1%]
3	Kimi Antonelli	P1	P5	18.4%	7.2%	20.6%	12.8%	[14.2–27.2%]
4	Charles Leclerc	P4	P2	5.9%	19.2%	7.1%	12.5%	[3.5–11.7%]
5	Oscar Piastri	P5	P6	7.9%	8.7%	3.2%	7.1%	[0.9–6.6%]
6	Lando Norris	P6	P4	0.7%	13.6%	2.0%	7.0%	[0.4–4.7%]
7	Liam Lawson	P12	P7	0.0%	5.4%	0.1%	2.4%	—
8	Max Verstappen	P8	P9	1.0%	4.1%	0.4%	2.4%	[0.0–1.6%]
9	Ollie Bearman	P10	P8	0.0%	4.7%	0.1%	2.1%	—
10	Esteban Ocon	P13	P10	0.0%	2.3%	0.0%	1.0%	—
11	Isack Hadjar	P9	P11	1.6%	0.9%	0.1%	0.9%	—
12	Pierre Gasly	P7	P12	0.1%	1.0%	0.1%	0.5%	—
13	Carlos Sainz	P16	P13	0.0%	0.8%	0.0%	0.4%	—
14	Nico Hülkenberg	P11	DNF	0.0%	0.7%	0.0%	0.3%	—
15	Alex Albon	P17	P14	0.0%	0.4%	0.0%	0.2%	—
16	G. Bortoleto	P14	P15	0.0%	0.4%	0.0%	0.2%	—
17	Fernando Alonso	P18	P17	0.0%	0.3%	0.0%	0.1%	—
18	Valtteri Bottas	P15	DNF	0.0%	0.3%	0.0%	0.1%	—
19	Lance Stroll	P19	P16	0.0%	0.2%	0.0%	0.1%	—
20	Arvid Lindblad	P20	DNF	0.0%	0.1%	0.0%	0.0%	—
21	Jack Doohan	P21	P18	0.0%	0.0%	0.0%	0.0%	—
22	Oliver Crawford	P22	P19	0.0%	0.0%	0.0%	0.0%	—

TYRE STRATEGY SIMULATION — TOP 5 DRIVERS

Each of the 5 spaghetti traces represents a different race simulation. The wide spread shows strategic variance — safety car timing shifts optimal stop windows by ± 4 laps. Hamilton's traces show noticeably steeper slopes from lap 20+ in many simulations, reflecting the modelled tyre degradation from his aggressive sprint driving style.

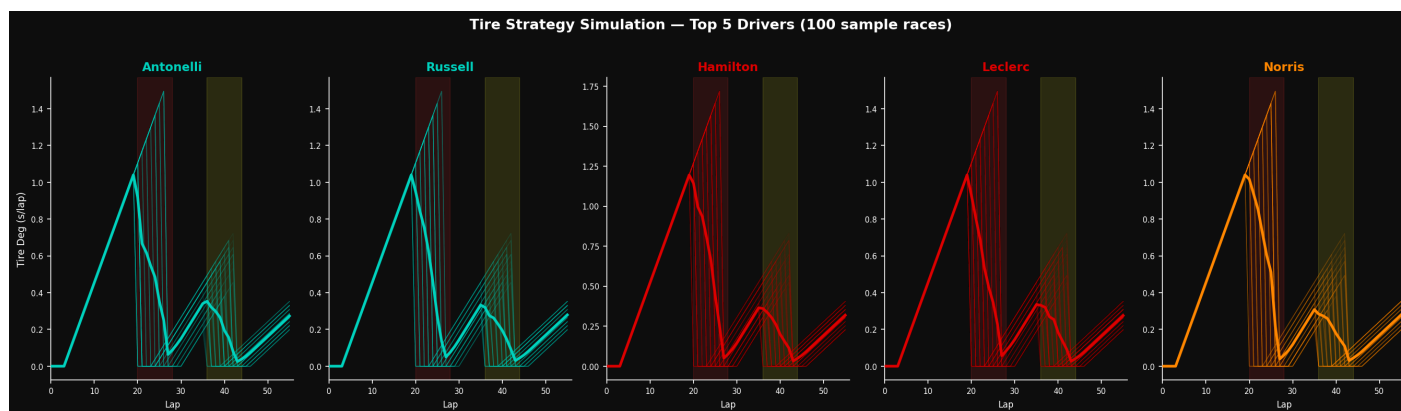


Chart 5 — 100 sample tyre strategy simulations per driver. Red/yellow shaded bands = optimal pit stop windows (stop 1 / stop 2).

RACE SCENARIO MATRIX — DRY/WET × SAFETY CAR/NO SAFETY CAR

The 2×2 scenario matrix is one of the most useful outputs — it shows where probability mass shifts under different race conditions. Safety cars hurt the leader (Russell) and help mid-grid runners. Rain is Hamilton's and Verstappen's best friend, and a potential disaster for a 19-year-old leading on slicks for the first time.

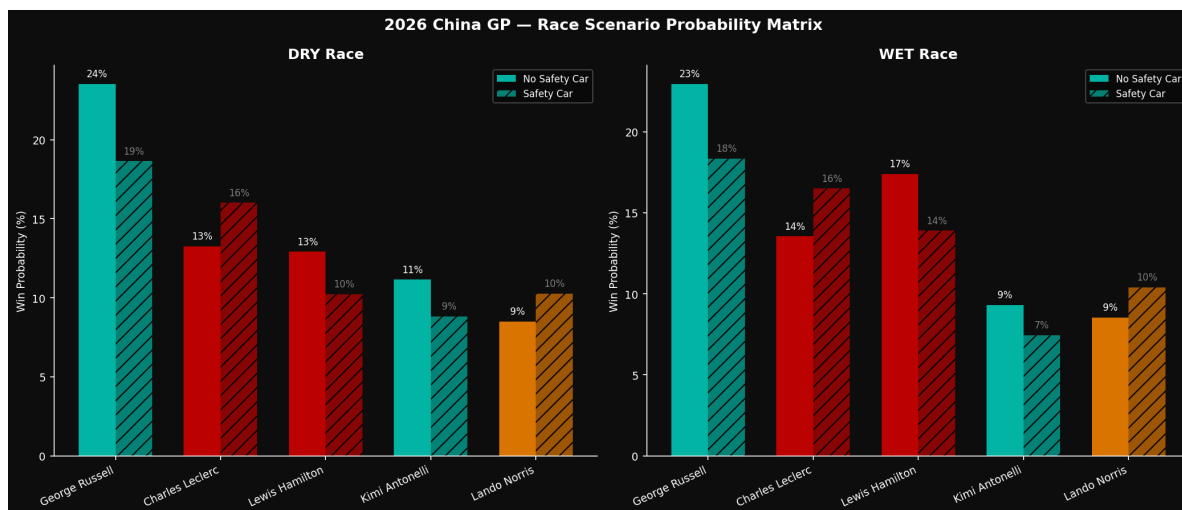


Chart 6 — Scenario matrix: solid bars = no safety car, hatched = safety car deployed. Left = dry, right = wet conditions.

CHAMPIONSHIP IMPLICATIONS — AFTER PREDICTED RESULT

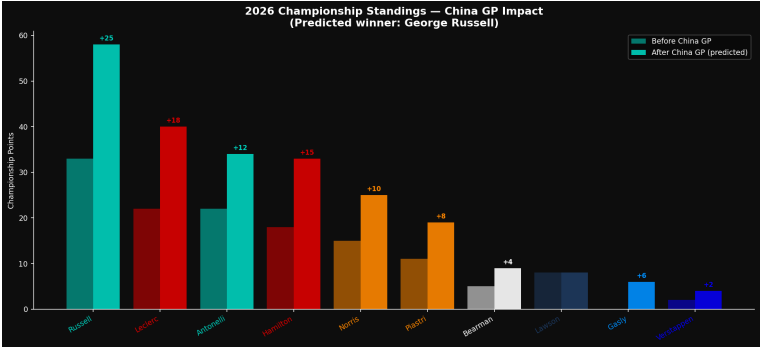


Chart 7 — Projected standings after China GP. Russell extends lead with predicted win. Hamilton closes gap via podium finish.

SPRINT RACE RESULTS (19 LAPS)

Pos	Driver	Gap	Note
1	George Russell	+0.000s	Wire-to-wire winner
2	Charles Leclerc	+0.674s	Charged from P6 — fastest lap
3	Lewis Hamilton	+2.554s	Briefly led lap 3
4	Lando Norris	+4.433s	Solid, unflustered
5	Kimi Antonelli	+5.688s	+10s penalty, true pace was P2
6	Oscar Piastri	+6.809s	Consistent
7	Liam Lawson	+10.900s	Best of the rest
8	Ollie Bearman	+11.271s	Strong sprint debut
9	Max Verstappen	+11.619s	Fell to P16, recovered
10	Esteban Ocon	+13.887s	Clean race
DNF	Nico Hülkenberg	—	Triggered safety car

V1 (AUSTRALIA) → V2 (CHINA)

Component	V1	V2 Change
Sprint Tier 0	None	Added — highest weight tier
DNF Multiplier	2×	Upgraded to 3× (Year-1 regs)
FP Proxy	FP2 missing	FP1 lap delta hardcoded
FP-Quali Divergence	Not encoded	fp1_to_quali_divergence
Hamilton P(win)	2.5%	21.2% — V1 failure fixed
MC Ensemble Weight	35%	45% (sprint calibration)
Bayesian Stages	1 (prior)	3 stages: prior+sprint+grid
Weather Model	None	25% rain blend added

KEY RISK FACTORS

Level	Risk	Probability	Who It Affects	Model Response
CRITICAL	Antonelli rookie pressure	Always present	Antonelli (P1)	22% retire rate prior; ×0.92 reliability penalty
HIGH	Safety car deployment	56% probability	All — hurts leaders	SC model: +25% probability for P5–P10 field
HIGH	Rain onset during race	25% probability	Hamilton +45%	Wet ensemble branch blended at 25% weight
HIGH	Russell Q3 reliability	Unresolved	Russell	×0.92 post-ensemble penalty + DNF flag
MEDIUM	Ferrari pit strategy risk	20% chance	Hamilton, Leclerc	Ferrari pit: 20% suboptimal stop in MC
MEDIUM	Hamilton tyre management	Sprint-confirmed	Hamilton (P3)	15% higher deg rate + pace ceiling laps 30+
LOW	Wind gusts (36 km/h)	Forecast	Turn 1 field wide	Not modelled directly — risk flag only

FINAL CALL: George Russell 28.6% · Lewis Hamilton 21.2% · Kimi Antonelli 12.8% · Charles Leclerc 12.5%
The race is Russell's to lose. Leclerc is the wildcard. If it rains, back Hamilton. If there's an early safety car, watch Lawson and Bearman.